

7. Integration Techniques

7.7. Approximate Integration

7.7 Learning Objectives

- I understand how to use constant, linear, and quadratic functions to approximate integrals.
- I understand the geometric interpretation of Simpson's Rule.
- I understand how the type of approximation used affects the accuracy of our answer.

7.7.1. Motivation

The fundamental theorem of calculus guarantees that any continuous function has an antiderivative. And we have talked a **lot** about how to compute antiderivatives and integrals by hand. However, it turns out there are even more antiderivatives that we simply can't write in terms of functions we know even if they exist in theory. This isn't always just a case of "we haven't tried the right technique," it's just that there are some continuous functions that have antiderivatives, and those antiderivatives just aren't any nice named function we have. And these come up a lot!

Unfortunately, integrals and antiderivatives have a lot of applications (some we have seen, some we will see soon, and more you may encounter in the wild) that are really important to all sorts of modeling, engineering, and other science applications.

Take for example

$$\int e^{x^2} dx \quad \text{and} \quad \int \sqrt{1+x^3} dx.$$

We cannot write these antiderivatives in terms of functions we know. But, from the FTC, we know the antiderivatives exist because $\int_a^x dt^2$ and $\int_a^x \sqrt{1+t^3} dt$ have continuous integrands.

Note

The way I talk about this section may make you think it is less important. It is sort of a tricky thing to test on in a timed environment without a calculator. However, this is one of the biggest sections to pay attention to and take away from this class if you are any kind of engineer, computer scientist, or other STEM major. There are full classes based on just the topics in this section, and it is an incredibly important field. Being able to approximate integrals quickly and efficiently is very important.

7.7.2. Recall Left/Right Riemann Sums

Forget any limits for now, we will eventually just use large enough n for our approximations. Then, to approximate integrals we have seen left and right endpoint approximations:

Definition 7.7.1 Left Endpoint Approximation

We approximate the integral by dividing the interval $[a, b]$ into n equally spaced rectangles, with width $\Delta x = \frac{b-a}{n}$. Then the height of these rectangles is given by the function value at the left endpoint of the rectangle.

$$\int_a^b f(x) \, dx \approx L_n = \sum_{i=0}^{n-1} f(x_i) \Delta x$$

Note, another way to write this sum is

$$L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x$$

it just depends if you are okay counting the first rectangle as rectangle 0 or 1.

Definition 7.7.2 Right Endpoint Approximation

We approximate the integral by dividing the interval $[a, b]$ into n equally spaced rectangles, with width Δx . Then the height of these rectangles is given by the function value at the right endpoint of the rectangle.

$$\int_a^b f(x) \, dx \approx R_n = \sum_{i=1}^n f(x_i) \Delta x$$

Observe if f is **strictly increasing** (always increasing, never decreasing) then L_n is always an underestimate and R_n is an overestimate.

Alternatively if f is **strictly decreasing** (always decreasing, never increasing) then L_n is an overestimate and R_n is an underestimate.

Exercise 7.7.1

Draw a picture to show the above claims, and convince yourself they are true.

In other cases of f , there is no guarantee if L_n or R_n are over or underestimates without further information.

Exercise 7.7.2

Can you think of (draw) an example where f is not strictly increasing nor strictly decreasing and L_n is an overestimate still?

7.7.3. More Approximations

When we talk about Riemann sums, we mention that the x_i^* can be any point in the sub-interval. The other point we will look at is midpoint rule.

Definition 7.7.3 Midpoint Approximation

We approximate the integral by dividing the interval $[a, b]$ into n equally spaced rectangles, with width Δx . Then the height of these rectangles is given by the function value at the midpoint between the left and right endpoints of the rectangle. In other words we use

$$x_{i+\frac{1}{2}} = \frac{x_i + x_{i-1}}{2}$$

the average of the left/right endpoints. Then,

$$\int_a^b f(x) dx \approx M_n = \sum_{i=1}^n f(x_{i+\frac{1}{2}}) \Delta x$$

We can get a bit of a better approximation if we use different shapes other than rectangles. In the next approximation, we will use trapezoids

Definition 7.7.4 Trapezoidal Approximation

We approximate the integral by dividing the interval $[a, b]$ into n equally spaced **trapezoids**, with width Δx . Then the heights of these trapezoids are given by the function value at the endpoints of the sub-interval. Then, the area of one rectangle is

$$A_i = \frac{\Delta x}{2}(f(x_{i-1}) + f(x_i))$$

The “middle trapezoids” share heights so that we get a bit of duplicated information except the left and rightmost trapezoids’ heights. Then,

$$\int_a^b f(x) dx \approx T_n = \frac{\Delta x}{2}(f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n))$$

Note, for f strictly increasing or strictly decreasing, we cannot conclude if the midpoint or trapezoid rules are over or underestimates. We need more information to conclude anything.

If f is **concave up** on $[a, b]$ then T_n is an overestimate, and M_n is an underestimate. If f is **concave down** on $[a, b]$ then T_n is an underestimate, and M_n is an overestimate.

Exercise 7.7.3

Draw a picture to show the above claims, and convince yourself they are true.

Note

T_n is our first time seeing an approximation that isn't made up of rectangles. But it turns out T_n is related to R_n and L_n . Observe we can rearrange the sum to see

$$\begin{aligned}
T_n &= \frac{\Delta x}{2}((f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n))) \\
&= \frac{\Delta x}{2}[(f(x_0) + f(x_1) + f(x_2) + \dots + f(x_{n-1})) + (f(x_1) + f(x_2) + \dots + f(x_{n-1}) + f(x_n))] \\
&= \frac{1}{2}[\Delta x(f(x_0) + f(x_1) + f(x_2) + \dots + f(x_{n-1})) + \Delta x(f(x_1) + f(x_2) + \dots + f(x_{n-1}) + f(x_n))] \\
&= \frac{1}{2}(L_n + R_n)
\end{aligned}$$

So the trapezoid rule is the same, numerically, as the average of the left and right approximations!

Example 7.7.1

Use M_5 and T_5 to approximate $\int_1^2 (\sin(e^x) + 2) dx$. Use an excel sheet to do so!

Solution

What do we need for Excel?

For M_5 we need to have

- $M_5 = \sum_{i=1}^n f(x_{i+\frac{1}{2}}) \Delta x$
- $n = 5$
- $\Delta x = \frac{2-1}{5} = \frac{1}{5}$
- $x_i = 1 + i(\frac{1}{5})$
- $x_{i+\frac{1}{2}} = \frac{1}{2}(x_i + x_{i+1})$

And for T_5 we need to have

- $T_5 = \frac{1}{2}(L_5 + R_5)$
- $L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x$
- $R_n = \sum_{i=1}^n f(x_i) \Delta x$
- $n = 5$
- $\Delta x = \frac{2-1}{5} = \frac{1}{5}$
- $x_i = 1 + i(\frac{1}{5})$

Then we set up in Excel

variable	value
a	1
b	2
n	5
deltax	0.2

and get

Variable	x	$f(x)$
x0	1	2.410781291

Variable	x	$f(x)$
x0.5	1.1	2.136994463
x1	1.2	1.822422509
x1.5	1.3	1.496448982
x2	1.4	1.208287429
x2.5	1.5	1.026493407
x3	1.6	1.028815175
x3.5	1.7	1.276238492
x4	1.8	1.768579232
x4.5	1.9	2.391912189
x5	2	2.893854955

See Excel file in Canvas Files for details and formulas. $M_5 = 1.665617507$, $T_5 = 1.696084493$

7.7.4. Simpson's Rule

There are many, many more complicated approximations for integrals. In this class, we will look into only one more, called Simpson's rule.

Consider that Trapezoid rule is kind of like treating the function like it is just a line on each sub-interval. To write the equation of a line, recall we need just 2 points it passes through. If we want to use a **higher order approximation**, we would need 3 points and those points would uniquely define a parabola. Skipping a lot of steps, we can say this next approximation method is like if we treated each consecutive pair of sub-intervals using a parabola.

Definition 7.7.5 Simpson's Rule

If we divide our interval $[a, b]$ into an even number of sub-intervals n of equal length $\Delta x = \frac{b-a}{n}$, then each consecutive pair of intervals can be approximated using a parabola. The resulting approximation will be given by:

$$S_n = \frac{\Delta x}{3} [(f(x_0) + 4f(x_1) + f(x_2)) + (f(x_2) + 4f(x_3) + f(x_4)) + \dots + (f(x_{n-2}) + 4f(x_{n-1}) + f(x_n))]$$

or

$$S_n = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$

Example 7.7.2

Use S_6 to approximate $\int_1^2 (\sin(e^x) + 2) dx$.

Solution

We will use a calculator or spreadsheet app again. Then we end up setting up the variables as

variable	value
a	1
b	2
n	6
deltax	1/6

and we get

Variable	x	$f(x)$
x0	1	2.410781291
x1	1.166666667	1.930378478
x2	1.333333333	1.393162833
x3	1.5	1.026493407
x4	1.666666667	1.164690632
x5	1.833333333	1.971519496
x6	2	2.893854955

See Excel file in Canvas for details and formulas. Then $S_6 = 1.674106039$.

7.7 Section Summary:

- We used Left/Right Riemann sums to approximate integrals.
- We used Trapezoid rule and midpoint rule to approximate integrals.
- We can discuss when left/right/trapezoid/midpoint rules are over/underestimates.
- We looked at a more complex approximation method, called Simpson's rule.